

Octopus: "Cognitive Bifurcations and Mathematical Explorations"

37:43

abstract for this talk was and I had to look this up this morning uh thinkers and thoughts have a co-evolutionary relationship this relationship is especially clear in the arena of mathematics where mathematicians forage for mathematical statements and structures in a process governed by complex and potentially unstable rules we will give a partial mapping of the structure of this space which necessarily relies not only on rigorous mathematical logic but also on subject beliefs metacognitive juristic and creative play so I've done a magic trick here all of these things are claims that are in need of evidence from my perspective and the goal here will be to provide some evidence in that direction but also to engage in conversation so in the construction of expertise I have no ambition here to play the role of expert there are certainly qualifications and evidence that I could point to my experience as a math professor for seven and half years my lived experience as a neurodivergent individual for 42 years my time spent in an octopus identity for the last three years openly and none of those things makes what I say here more valuable than what you think about them the only request I will make is that I've given a lot of effort into thinking through these thoughts some of which were quite difficult that showed

up for me and I ask that you will give some level of effort towards finding the value in those thoughts and how they can be mapped to your reality so let's move from the abstract to abstraction when people think of mathematics in the modern sense they often begin here as both its value and its difficulty we might imagine that Bob and Alice both have some cows Bob has two and Alice has three and they're curious how many cows they would have if they combine their herd Alice can bring her cows to Bob and they can count them then if they have the physical process of counting at their disposal a cognitive tool of one-to-one mapping between objects and objects but they can also do it conceptually one of the things that abstraction enables is the ability to move information about physical perceptions more quickly than the physical things themselves can move in other words Alice doesn't have to bring her cows with her if she can bring the concept of three or the idea that she has three so this is another aspect of mathematics that is often highly valued and highly contentious in relation to the ways in which we realize and reify scientific practice education and other ways in which education other ways in which mathematics shows up in our lives is how are we actually giving a physical body to the concept that we are hoping to

represent the fact that two apples are
con conceptually in some sense the same
as two cows or two
fingers is one level a level above that
is the fact that we can attribute
symbols to these
representations anything from two
interlaced eyes and the Roman numeral
setting to two in the Hindu Arabic
numeral setting to words like two and do
to two dots to two
fingers the two fingers are physical yet
somehow they're also a basis of
abstraction
because the idea of abstraction includes
the fact that there must be something
concrete to
abstract and this two fingers I've
included on the page of abstract
representations partially because I like
the idea of things having bodies I like
the idea of the number two having lots
of different bodies how many bodies can
one idea have but also I like the
ambiguity here between the two and the
three because in some systems the way
that I'm showing this here would be
interpreted as three and in others as
two so with that ambiguity in mind there
is no slide
three please do not discuss slide three
later we will not bring it up
again and in fact it is not advantageous
to you to do so because the claim that
there is no slide three is an idea
wholly owned by eight arms nine brains
and if you want to discuss the fact that
there is no slide three

there will be a licensing fee
involved thankfully some ideas are open
to discussion with no
um financial transaction involved and
one of them as we finally come to slide
four is a very important Concept in
mathematical pedagogy called the rule of
four I think this encapsulates thinking
that might have been going on for a
while before it was formally published
but the first time I encountered it was
in the excellent calculus textbook by
Hughes Howen gleon and McCullum at all
this was the first calculus textbook I
looked at When I Was An undergraduate
student and it was also the first
calculus textbook I looked at when I was
a beginning Professor beginning to teach
my first Calculus class I'm afraid that
far too many students probably
completely skip this section uh which is
in the pre Preamble uh but one of the
key Concepts that it elucidates is how
to think about mathematics
and it's not saying this is the right
way it's saying this is a way and this
is a way that the textbook uses quite
profitably it's called the rule of four
and it says essentially that when you
encounter a new mathematical concept
there are four distinct but interrelated
ways that one might think about it you
could give a verbal description or a
definition um you could draw a graph or
some type of diagram you could use
numbers in the sense of
data or you you could use a symbolic
representation like a formula or an

equation and the extent to which these are distinct to inter related highlights the value that each of these may have to individuals as a latching on point for thinking about difficult mathematical Concepts and calculus it also eliminates some of the privilege that might exist around specific representations around the idea that symbols might be real math whereas when we explain them at a high level somehow that isn't real math for instance if we look at the types of mathematical works that mathematicians actually produce um it might be surprising to undergraduate students what percentage of this is not what they would traditionally think of as symbolic and what I would mean here is equations and formulas like we get to page three and finally we see something that looks like a symbolic mathematical expression but the majority of the work here is spent on words words and diagrams visuals a large portion of this is in the theorems of course which are something that is uh a very individual feature of pure mathematics but let's not get bogged down there let's simply say there are many different representations and one of the challenges of an effective instructor is to map between these different representations so that people who might feel most at home in one of them can use that as a leverage point for accessing the other one

uh based on my background I'm I'm a
defro math professor voluntarily also a
chaos magician and so I'm very very
tempted to try to build a funter between
this rule of four originating in
mathematics and this four quadrants
model of Human Experience Advanced by
Ramsey Dukes in his book known as s
sopy um which says that there are these
four lenses through which human beings
can process and structure experience
there's art religion magic and science
it's so tempting to try to do something
like say the symbols are the magic and
the words are the religion and the
numbers are the science and the graph is
the art and I'm going to resist that
Temptation because I think it's it might
be a useful initial temptation but the
resistance of the Temptation provides
its value instead I'm going to hold this
as an idea

we will return to it in a moment where
it will play against some other Concepts
that we're likely to
encounter I will stop here for a moment
and say something in promptu though
which is

um there's a practice in this world of
chaos magic called belief
shifting where the magician consciously
and intentionally adopts a specific
belief as if it were something they
believe to be true and there's a very
related practice in mathematics which is
the construction of the
axioms or the adoption of the axioms so
in mathematical proofs once you adopt

specific axioms or specific assumptions
the conclusions should follow not
without effort but in a clear logical
flow mathematicians are capable of
adopting different
axioms and this produces what I'm going
to call a cognitive bifurcation
so a cognitive bifurcation since we've
already split things into four they may
be too much let's try to get things back
together and just split it into
two one place that a cognitive
bifurcation might show up in the history
of mathematics is in the different types
of geometry that geometers think about
so one of the features of modern
geometry is the realization that the
parallel postulate in Euclidean Geometry is
not possible to prove from the other
axioms of Euclidean
geometry and so the parallel postulate
says that given a line and a point not
on that line there is exactly one
parallel line or there's actually one
line that passes through that point
parallel to the original
line and if you choose not to adopt this
belief as Euclid played with in the idea of
one of his lectures um it opens up a
whole new world of non-Euclidean geometry
both spherical and
hyperbolic so the axioms that we adopt
and the beliefs that we intentionally
choose to consider are powerful tools
that can open up new conceptual
playgrounds but also quite
dangerous Alice and Bob probably feel
very safe in this world even if we don't

admit a slide three we admit a number
three to describe how many cows one of
them may
have and then the question is are these
all the numbers and and if they are not
in which directions would we be willing
to consider perhaps we asking what comes
after five or after the number after
five and like a child we realize that we
can continually add one and there is no
a priority upper bound and we come
across the idea of
infinity perhaps we start to look for
what could be in between the numberers
and so we discover numbers like one and
a half or
1.1 or perhaps we make a dangerous leap
in the other direction and we admit the
possibility that there is a number which
says that you have no
cows and while you might need such a
number is an intriguing question but
let's say that we do have one when we
introduce a number in this way we need
not only to consider the fact that the
number exists we need to consider the
ways in which it will interact with the
other numbers that exist in other words
the rules so as we begin to if we want
to we know how to add numbers that were
one through five but how do we add to
zero and this is a cognitive bifurcation
because now we have a symbol that plays
by its own rules and we've opened up a
whole new world of numbers to where when
we added positive numbers together they
always increased we were always heading
to the right the fact that we can now

introduce a number where we add it and identity for the binary operation of addition is uh an interesting an interesting split that opens up more possibilities on the left and side and I would say this is a cognitive bifurcation it says in the the definition that I'm adapting from Wikipedia a bifurcation occurs when a small smooth change made to the parameter values of a system causes a sudden qualitative or topological change in its Behavior it's hard to imagine a change smaller or smoother than putting the circle here and declaring that it doesn't do anything on addition and yet this an immense cultural controversy when these numbers are being introduced so I'm going to make the claim that mathematical thinking systems are complex systems I'm going to give the highly unsatisfactory proof by Authority that this claim is obvious one piece of evidence we could point to is that the existence of a parameter bifurcation almost immediately suggest that we're dealing with a complex system and yet that is not quite the right way to think about it especially if we're going to be looking at it from the active inference context so let's ask for more evidence and thus not accept proof by Authority what evidence do we see for the belief that mathematical thinking systems are complex systems and one way that we might uh

conceptualize this through a model is at the agent level where we think about the types of agents who are involved in these

systems I've heard that there would be people here who like ants so I've made sure to include some and uh here's my aunt Hill and here are my ants and I have these ants foraging for formulas that they bring back to the colony which then feed other ants which are a part of the ecosystem the formulas and the symbols that they bring back in this mathematical world of semantics is more than just symbols right it's something that actually contributes to the survival of the colony but in order to do that it must be converted into other

things so so it would be easy to just try to be efficient through this and reduce it as much as we could to something useful I'm sympathetic to that Viewpoint because mathematics has to be useful at some level to human existence to justify its existence I

believe so if we were looking at this on a belief level on a on a probability level we might say something like with certain axioms as priors mathematical theorems are simply

beliefs such that an agent in order to survive who quickly will quickly adopt correct beliefs with probability one if you want to make $2 + 3 = 6$ you will quickly run into issues in which everything collapses and it ceases to be a useful

tool and and I'm not opposed to this model I just want to say it's oversimplified because many of the beliefs that mathematicians choose to adopt and choose to spend their valuable resources thinking about don't have any immediate mapping to applications for instance the question of whether or not zero is a natural number is something mathematicians will argue about what is the utility in foraging for this belief perhaps you can think of one um but I think it speaks to something else that could be happening that mathematicians throughout the centuries have seen which is that the the claims of truth that you make are not entirely limited by the axioms that you assume and by the process of logic there's there are larger questions here cultural questions uh of identity ontological questions of what is real what does it mean for zero being a natural number to be a true statement mathematicians find it useful to disagree about this the conversations that we have about these things make us better mathematicians I feel I'll never forget when my PhD adviser explained to me why he believed that zero was a natural number but there's no mathematical proof that can provide this it's an axiom that you accept or do not when you define these numbers and when we talk about types of numbers there's a famous cognitive vacation that occurred at some point in history which is the division uh the

discovery of the fact that not all
numbers are rationals not all numbers
can be expressed as a ratio between two
integers

if you're not willing to accept this
then you cannot come up with a solution
to $x^2 - 2 = 0$ and of course you could
by authoritarian fiat simply declare
that there is no solution to $x^2 - 2 = 0$

[Music]

There's a legend that the
mathematician Hippasus who discovered
that the square root of two was
irrational was thrown overboard by
the Pythagoreans so that he would not
share this information and I am imagining
this as a kind of cyberpunk reimagining
of ancient Greek geometry not as an
insult in any way to Pythagoras who I
think would appreciate this but to
imagine that there is a geometer who is
not opposed to the idea but opposed to
the pathway by which it came through and
willing to use Force to ensure that that
dangerous doorway stays
closed so another model of math person
perhaps that we could view is as a
boundary between wherever math comes
from and the unquote real
world and the real world is in full of
Agents already that we can easily map to
there are institutions educational
institutions governmental institutions
um professional societies there's
politics where the government may decide
certain levels of funding for those
institutions business which may choose

to fund the institutions to uh insert
itself into the process of politics
there are funding sources there's media
one thing that occurred to me as I Was
preparing this talk is we don't often
think of mathematical journals or
scientific journals as media but they
are they're media that's made and
constructed like any
other and then is the question of where
math comes from where does math come
from
uh this is what I've wrestled with a
while one reason is it opens up some
very
some doors that from the material
materialist utilitarian perspective of
like a School of Engineering feel really
strange where did Alice and Bob first
get that idea of numbers and does doing
math involve some type of contact with
these strange things that could
happen we don't see very much of this in
our schooling which is often about
producing symbols uh producing people
who can follow precise rules with
respect to symbolic manipulation and at
that point it can become a type of
overcorrection and I'm sympathetic to
this door that uh I'm sympathetic to
this perspective that opens the door a
bit more to the strange not only by my
own temperament but also through the
lived experience of people who
accomplished very effective physical and
Engineering uses of mathematics but were
willing to express that they were
sympathetic to this viewpoint so Hinrich

Herz For Whom the the unit of a cyclic
repetition and a computer processing
unit is named for says one cannot escape
the feeling that these mathematical
formulas have an independent existence
and in intelligence of their own that
they are wiser than we are wiser even
than their discoverers in other words if
the intelligence is in the symbols and
not in the humans who are trying to work
with them as tools
this provides a very different
perspective on the creation of
mathematical thought and on the ways in
which mathematical thought not only
originates but also can be
applied it also points out to some
extent the horror that we can be doing
when we do this
non-consensual cognitive
bifurcation of a person of people
persons into math persons and not math
persons Often by their ability to match
some preconceived rule to arrive at a
correct answer often with an
overemphasis on
speed so as we think about the fact that
the people the persons the
mathematicians the math persons may be
foraging for these symbols and formulas
to bring back to the real world it's
what happens if we begin to
give if we as hers it begin to give
these symbols their own ontology and
their own agency to realize that they
are also likely out in the wild
searching for the people to whom with
whom they will engage

productively so this takes us back to
our stranger ideas where we
intentionally left the door open and
here I'm indebted to Gard mayor who in
his article magicians of the 21st
century um pointed to Ramsey dukes's for
quadrants model as splitting reality
into the irrational and the rational
intentionally left the rational box off
here because I wasn't sure where to put
it I would put religion in the rational
category because there are very clear
apologetic logical arguments it's clear
that religion at least attempts to play
in the same sphere of the rational as
science does whereas art and Magic would
be in the irrational category and I am
intentionally echoing the splitting here
of the real numbers into rational
irrational with whatever consequences
this might have the quote hinr Hertz
again outside our Consciousness there
lies the cold and alien world of actual
things between the two stretches the
narrow Borderland of the senses no
communication between the two worlds is
possible excepting across the narrow
strip for a proper understanding of
ourselves and of the world it is of the
highest importance that this Borderland
should be thoroughly
explored and the problem as map makers
and an Ever evolving and somewh say
chaotic territory certainly highly
Dynamic of this moment and SpaceTime
where we find ourselves is where to draw
where is the border if this is the
border that we're supposed to be

exploring for a proper understanding of ourselves and of the world how do we know when we are on the border and when we may be inadvertently Crossing it and how can we enrich our tools for finding each side and productively bringing ideas from one side to the other or whatever Travelers are wishing to cross whatever we call

them I've late recently been reading this book symbolism by Alfred North Whitehead and he speaks to this but he speaks to it in a different way and in one that I would challenge which is very intimidating to be honest um but in this in this essay on symbolism he talks about essentially two types of symbolism the symbolism of the church and the symbolism of language and of

algebra so he initially says that there is a symbolism of the medieval cathedral but because that symbolism can be dispensed with it is on The Fringe of life and has an unessential element in its

Constitution and then he says and on page two

here there is another sort of language a written language which is constituted by the mathematical symbols of the science algebra in some ways these symbols are different to those of ordinary language because the manipulation of the algebraic symbols does your reasoning for you provided that you keep to the out algebraic

rules how can something do reasoning for you if it is purely mechanistic and has no agency and then he says this is not the case with ordinary language which puts up that boundary again between words and symbols that the symbols can do something the words cannot so I would disagree here with how he draws the bifurcation between Cathedrals and mathematical symbols uh Pythagoras would also have quibbled with that pythagora for the pythagoreans mathematics was very much a religious practice and they believed that numbers had a spiritual Dimension but that takes us to a very weird that takes us to very weird places very quickly uh I tweeted this I don't necessarily believe it but it's you don't have to tweet everything you believe or vice versa uh so let's go to slightly a bit more out there but this is coming back to relevance I promise it has to do with the notion of sovereignty and so here I'm looking at the article sovereignty in the UFO by win and Duvall where they talk about how sovereignty is constructed and so this might go back to our definition here of a quote unquote real world an institution a government a nation state even an individual as being Sovereign has a a particular Construction in the world of our experience but this has not always been the case I loved this quote that they said anthropos anthropocentric

sovereignty might seem necessary after
all who else besides humans might rule
nevertheless sovereignty was LE
historically sovereignty was less
anthropocentric for Millennia nature and
the gods which nature would be in
science and the Gods be and religion
were thought to have causal powers and
subjectivities that enable them to share
sovereignty with humans if not exercise
Dominion outright authoritative belief
in non-human sovereignties was given up
only after long and bitter struggle
about the borders of the social world so
again we're trying to place a border in
which who what could be Sovereign
depends on who what should be included
in
society and there's a great footnote
here
anthropocentrism need not mean all human
beings since historically many physical
humans were not considered human
socially and let me go back here and say
that perhaps there's a similar
construction
occurring perhaps this division some
humans are human and some are not some
we would not recognize that today but we
do recognize bation in the math persons
and non-math persons and I would say
it's equally as social and equally as
artificial and equally as
dangerous so where are we now U so I've
opened some doors here perhaps some of
them should have stayed shut but
regardless they're open for the moment
and we can shut them when we wish

hopefully I love this quote at the end
about sovereignty in the UFO the UFO can
be known only by not asking what it is
uh because this has been totally obverse
to my process here I have tried to know
mathematics by asking what it is by
struggling very hard with its identity
as an egregor as a
concept um but maybe some things can
only be known by not asking what they
are so I'm sure that this has perhaps
been an interesting metaphysical IAL
exploration of some different historical
ideas and ideas of History how is this
applicable to ourselves as scholars in
an active inference
environment well what I would like to
point at is the ways in which some of
these ideas that are already Dynamic may
be changing very rapidly at this
moment so if we look at the identity of
mathematician if we are solely confined
to a meatspace identity that has
certain
connotations but what do we do with
someone like bluecow and I did request
blue Cal's consent to use their likeness
and their thoughts uh the creator of the
first hyperdimensional Computing OS and
the super prompt a prompt God uh someone
who is a self-proclaimed mathematician
who does impressive mathematical work in
prompting in prompt engineering of
mathematical prompts via GitHub and
whose appear par is a an
anime person dressed as a
cow um can they be a real mathematician
in the way that we

currently conceptualize math and are the mathematical ideas that they're bringing ones that we are prepared to reckon with as

math and then when they post things like this what do we do self-identified skitso and it's helpful here perhaps to remember that skitso means split another scholar in this emergent might one might call parallel or underground genre whose work I respect is Sir Loop and

Watson his conjecture soon you'll be able to buy an AI hacker for less than \$100,000 he'll or she do he he he'll or or she do anything for free you'll need to pay only once and it will last forever this is exactly the bargain of the

theorem once you have proven a theorem once with a finite amount of human effort it remains irrevocably true and will do whatever you ask it to do for free

forever so if we believe there is a world where theorems come from what does that say about other objects or other constructions that might share their properties

I don't have a conclusion if I were going to have a conclusion it would be it going back to the thoughts of slide three but one reason I don't have a conclusion is that this is not the end of a completed research Journey but the beginning of something we are all experiencing together so I'm gesturing at what I know and then the next

question is what do I hope to know well one challenge that I think this immediately offers but also an opportunity is what does it look like to be an institution in a world where these agents these human agents are seriously considering the same ideas as you are and perhaps offering insights that are not being produced at the level of your institution I don't this may be Fringe but I think it's to be taken seriously and I offer it because many people will see the work that for instance Terry tow is doing on leading larger mathematical teams that use AI to modularize math and prove more efficiently in an engineering type framework different from the idea of the individual math um it's less likely that the average academic I believe would encounter these types of people who are also doing interesting mathematical thinking not only about the mathematics itself but about what it means to present oneself as a mathematician or to present mathematical work so I hope these thoughts have been interesting to you and excited to see what they may have brought up in the form of questions and comments thank you ever so much octopus um I really appreciate the number of different ways that you've kind of viewed this topic um if I may start with with a question sure so what as you were talking about this idea that the formulas maybe exist and that they're kind of waiting to fit on

to something it was reminding me of a um
a song by the Icelandic uh artist Bor
I'm not sure if you're familiar with her
she has
a a song called modern things where it
says all those modern things they've
always existed they've just been waiting
in a mounting for the right moment um
and it's their time now and um so yeah I
was wondering you know what what do you
think
about um something like the global
economy which it's um it kind of acts
like an entity in itself so it's you
know it's the Invisible Hand of the
market for
example but it's the humans that kind of
assemble around it and imbue the meaning
on the all the uh interactions and all
Dynamics yes uh I think again that there
again there might be a border uh for
instance if we view economics as purely
human transactions of material Capital
it misses a lot I think a lot of
economists would say that but symbols
are important even there if you look at
a unit of currency it is imbued with
symbols that are highly
non-functional almost overridden with
nonfunctional and one of the ways that
one can assert a Authority and or
immortality is to issue a coin and
that's what we see now with
cryptoeconomic systems is a debate
around the individual sovereignty and
the right of who is allowed to issue
something that can be used as a unit of
exchange uh but that's not a new debate

there were certain Roman citizens who
had the authority to issue a
token and then the question is uh where
does the value of that token come from
ultimately right so so again it kicks
back very quickly to questions around uh
beliefs and identity and
sovereignty I guess the thing there is
when something is real to me it must
have more than just a material
component thank you um Daniel are there
any questions from the chat
there's a lot of great commentary
octopus it was a great presentation do
you have any final
comments oh um well I really appreciate
the invitation to be heard and the
ability to show up as octopus that's so
meaningful to me and U grateful to the
organizers and the audience and looking
forward to seeing what conversations can
emerge from this
beginning awesome
thank you till next time see you ooto
thank
you for